Views - Assignment 11

1) Create a view that shows all of the customers who have the highest ratings.

```
KD3-Prerana-96192>CREATE VIEW v1_create AS  SELECT * FROM customers WHERE rating =
    -> (SELECT MAX(rating) FROM customers);
Query OK, 0 rows affected (0.02 sec)

KD3-Prerana-96192>select * from v1_create;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.01 sec)
```

2) Create a view that shows the number of salespeople in each city.

```
KD3-Prerana-96192>create view v1_speopincity AS SELECT city, COUNT(*) AS sales_people_in_city FROM salespeople GROUP BY city ;
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>Select * FROM v1_speopincity;
+----------+---------------------+
| city     | sales_people_in_city |
+----------+---------------------+
| London   |                   2 |
| San Jose |                   1 |
| Barcelona |                  1 |
| New York |                   1 |
+----------+---------------------+
4 rows in set (0.00 sec)
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

```
KD3-Prerana-96192>CREATE VIEW v3_det AS SELECT  s.sname, count(o.onum) total, avg(o.a
mt) average FROM orders o INNER JOIN salespeople s ON s.snum = o.snum GROUP BY s.snam
e;
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>SELECT * FROM v3_det;
+---------+-------+-------------+
| sname   | total | average     |
+---------+-------+-------------+
| Rifkin  |     2 |  558.425000 |
| Peel    |     3 | 5127.356667 |
| Motika  |     1 | 1900.100000 |
| Serres  |     3 | 1848.716667 |
| Axelrod |     1 | 1713.230000 |
+---------+-------+-------------+
5 rows in set (0.00 sec)
```

4) Create a view that shows each salesperson with multiple customers..

```
KD3-Prerana-96192>CREATE VIEW Salespeople_With_Multiple_Customers AS SELECT s.snum, s.
sname, COUNT(c.cnum) AS total_customers FROM Salespeople s JOIN Customers c ON s.snum
= c.snum GROUP BY s.snum, s.sname HAVING COUNT(c.cnum) > 1;
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>SELECT * FROM Salespeople_With_Multiple_Customers;
+------+--------+----------------+
| snum | sname  | total_customers |
+------+--------+----------------+
| 1001 | Peel   |              2 |
| 1002 | Serres |              2 |
+------+--------+----------------+
2 rows in set (0.01 sec)
```

Changing Values through Views

5) Which of these views are updateable (will allow DML operations)?

#1. Create View Dailyorders as Select Distinct cnum, snum, onum, odate from Orders;

```
 at line 1
D3-Prerana-96192>SELECT * FROM  customers WHERE rating >= ANY(SELECT MAX(rating) FROM customers WHERE c
ty = 'Rome');
------+----------+----------+--------+------+
 cnum | cname    | city     | rating | snum |
------+----------+----------+--------+------+
 2002 | Giovanni | Rome     |    200 | 1003 |
 2003 | Liu      | San Jose |    200 | 1002 |
 2004 | Grass    | Berlin   |    300 | 1002 |
 2008 | Cisneros | San Jose |    300 | 1007 |
------+----------+----------+--------+------+
 rows in set (0.00 sec)

D3-Prerana-96192>CREATE VIEW Dailyorders AS SELECT Distinct cnum, onum, odate from Orders;
uery OK, 0 rows affected (0.01 sec)

D3-Prerana-96192>select  * from Dailyorders;
------+------+------------+
 cnum | onum | odate      |
------+------+------------+
 2008 | 3001 | 1990-10-03 |
 2001 | 3003 | 1990-10-03 |
 2007 | 3002 | 1990-10-03 |
 2003 | 3005 | 1990-10-03 |
 2008 | 3006 | 1990-10-03 |
 2002 | 3009 | 1990-10-04 |
 2004 | 3007 | 1990-10-04 |
 2006 | 3008 | 1990-10-04 |
 2004 | 3010 | 1990-10-04 |
 2006 | 3011 | 1990-10-04 |
------+------+------------+
0 rows in set (0.00 sec)

D3-Prerana-96192>INSERT INTO Dailyorders VALUES(2010, 3012, '1991-11-05');
RROR 1471 (HY000): The target table Dailyorders of the INSERT is not insertable-into
D3-Prerana-96192>
```

#2. Create View Custotals as Select cname, Sum (amt) Sum_Amt from Orders, Customers where Orders.cnum=Customers.cnum Group by cname;

```
KD3-Prerana-96192>CREATE VIEW Custotals as Select cname, Sum(amt) Sum_Amt from Orders, Customers Where O
rders.cnum = Customers.cnum Group by cname;
Query OK, 0 rows affected (0.02 sec)

KD3-Prerana-96192>select * from Custotals;
+----------+----------+
| cname    | Sum_Amt  |
+----------+----------+
| Cisneros |  1116.85 |
| Hoffman  |   767.19 |
| Pereira  |  1900.10 |
| Liu      |  5160.45 |
| Giovanni |  1713.23 |
| Grass    |   385.70 |
| Clemens  | 14614.88 |
+----------+----------+
7 rows in set (0.00 sec)

KD3-Prerana-96192> INSERT INTO Custotals VALUES('Ankit',15000.00);
ERROR 1471 (HY000): The target table Custotals of the INSERT is not insertable-into
KD3-Prerana-96192>
```

#3. Create view Thirdorders as Select * from Dailyorders where odate='1990-10-03';

```
KD3-Prerana-96192>Create view Thirdorderss
    -> as Select * from Dailyorders where
    -> odate= '1990-10-03';
Query OK, 0 rows affected (0.02 sec)

KD3-Prerana-96192>select  * from Thirdorderss;
+------+------+------------+
| cnum | onum | odate      |
+------+------+------------+
| 2008 | 3001 | 1990-10-03 |
| 2001 | 3003 | 1990-10-03 |
| 2007 | 3002 | 1990-10-03 |
| 2003 | 3005 | 1990-10-03 |
| 2008 | 3006 | 1990-10-03 |
+------+------+------------+
5 rows in set (0.01 sec)

KD3-Prerana-96192>UPDATE Custotalss SET Sum_Amt = Sum_Amt + 1000 WHERE cname = 'liu';
ERROR 1146 (42S02): Table 'sales.custotalss' doesn't exist
```

#4. Create view Nullcities as Select snum, sname, city from Salespeople where city is NULL OR sname BETWEEN 'A' and 'MZ';

```
KD3-Prerana-96192>Create view Nullcities
    -> as Select snum, sname, city
    -> from Salespeople
    -> where city is NULL
    -> OR sname BETWEEN 'A' and 'MZ';
Query OK, 0 rows affected (0.01 sec)

KD3-Prerana-96192>Select * FROM Nullcities;
+------+---------+----------+
| snum | sname   | city     |
+------+---------+----------+
| 1004 | Motika  | London   |
| 1003 | Axelrod | New York |
+------+---------+----------+
2 rows in set (0.00 sec)

KD3-Prerana-96192>INSERT INTO Nullcities VALUES(1005, 'Jully', 'Paris');
Query OK, 1 row affected (0.01 sec)
```

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

```
KD3-Prerana-96192>CREATE VIEW Commissions AS SELECT snum,comm FROM salespeople
    -> WHERE comm BETWEEN .10 AND .20 WITH CHECK OPTION;
Query OK, 0 rows affected (0.02 sec)

KD3-Prerana-96192>UPDATE Commissions SET comm = .30 WHERE snum = 1003;
ERROR 1369 (HY000): CHECK OPTION failed 'sales.commissions'
KD3-Prerana-96192>select * from Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.12 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.00 sec)

KD3-Prerana-96192>UPDATE Commissions SET comm = .30
    ->  WHERE snum = 1003;
ERROR 1369 (HY000): CHECK OPTION failed 'sales.commissions'
KD3-Prerana-96192>UPDATE Commissions SET comm = .12
    ->  WHERE snum = 1002;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```